SESSION 5

GeoAI Ethical Challenges & the Responsible Design Framework

Five Global Challenges · Identify–Assess–Mitigate–Monitor · Full Team Tropika Walkthrough · A Mitigation Toolbox · Ethical Risk Mapping

Day 1 · 7 July 2026 · GeoAI Hackathon Programme

How to use these notes. *These notes are self-contained: five documented ethical challenges, the full four-step design framework with diagrams, a complete step-by-step worked example, a practical mitigation toolbox, glossary, and self-check questions.*

Learning Objectives

By the end of this session, you should be able to:

- Describe five ethical challenges specific to GeoAI, each anchored in a real global case.
- Apply the four-step Responsible GeoAI Design Framework: Identify → Assess → Mitigate → Monitor.
- Prioritise ethical risks using a Likelihood × Severity grid under hackathon time constraints.
- Select appropriate mitigation techniques for bias, privacy, explainability, environmental-cost, and misinformation risks.
- Produce an Ethical Risk Register with impacts, mitigations, and owners (Activity 5).

1. Five GeoAI Ethical Challenges — with Real Cases

1.1 Data Bias

Underrepresented regions or groups in training data mean less accurate predictions exactly there — and in GeoAI, “there” is usually where vulnerability is highest, because poorer and rural areas are systematically less-mapped and less-sensored.

Real case: MIT Media Lab’s 2018 Gender Shades study tested commercial facial-recognition systems and found error rates of 0.8% for lighter-skinned men versus up to 34.7% for darker-skinned women — a 43× gap traced to unrepresentative training data. The same mechanism hits computer-vision models trained mostly on well-mapped, wealthy regions. The study’s impact was real: IBM withdrew its facial-recognition product entirely in 2020 (Buolamwini & Gebru, 2018).

1.2 Privacy & Surveillance

High-resolution imagery combined with AI enables monitoring at a scale and intrusiveness that neither could achieve alone.

Real case: Clearview AI scraped an estimated 30–50 billion photos from the open internet without consent, building a facial-recognition database sold to law enforcement including the FBI. Dutch, French, Italian, and Greek data-protection regulators have each fined the company tens of millions of euros — a reminder that “publicly visible” does not mean “free to harvest”.

1.3 Explainability

Many GeoAI models are “black boxes” offering limited interpretability — and a model can be right for entirely wrong reasons, which makes it fragile in exactly the situations that matter.

Real case: an AI model trained to detect skin cancer learned to associate a ruler in the photo with malignancy — dermatologists tended to place rulers next to lesions they already suspected were cancerous. The model scored well in testing while learning nothing about skin. The flaw went undetected until researchers checked what the model was actually looking at — the strongest practical argument for explainability tools like SHAP and saliency maps.

1.4 Environmental Cost

Training and running large models consumes serious energy. Context: training a large model can emit hundreds of tonnes of CO_2 against a global per-capita average of roughly 4.4 tonnes per year.

Real case: the International Energy Agency projects AI-driven data-centre electricity demand will roughly double by 2030; Google alone reported emissions rising nearly 50% over five years, driven largely by AI compute. For GeoAI specifically, the sustainability question is: does this model's value justify its footprint — and would a smaller model do?

1.5 Misinformation — "Deepfake Geography"

Manipulated or fabricated geospatial data can mislead decisions at scale — and maps carry an inherited authority that makes fake ones unusually persuasive.

Real case: University of Washington researchers demonstrated "deepfake geography" — using GANs (Generative Adversarial Networks, AI systems that generate realistic synthetic media) to create entirely fictional but convincing satellite images, blending real features of one city onto another. Security officials now flag manipulated satellite imagery as a serious threat (Zhao et al., Cartography & GIS, 2021). Practical defence: verify provenance and cross-check any imagery you did not source yourself against a second, independent source.

2. The Responsible GeoAI Design Framework

A four-step loop for building ethics into your project rather than bolting it on afterwards. What you learn in Monitor feeds straight back into a fresh Identify — the same way Learn feeds Observe in the Innovation Process (Session 2).

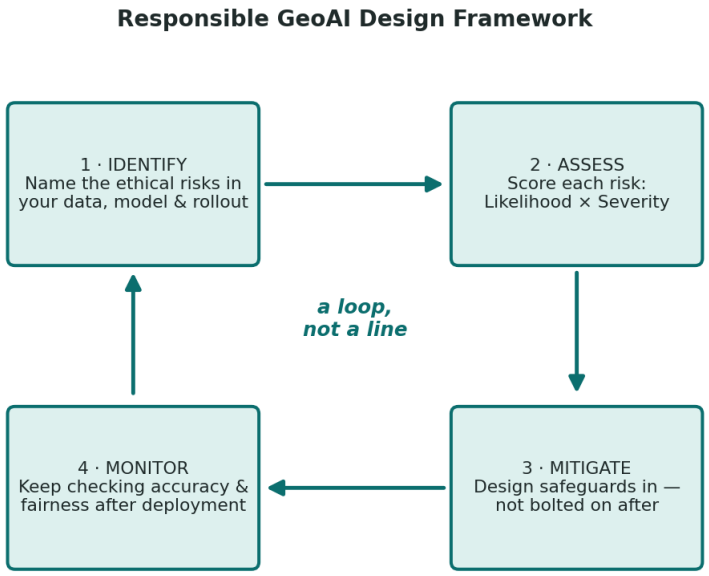


Figure 5.1 — The Identify → Assess → Mitigate → Monitor loop.

Step	What you do	Technique
1 · Identify	List where ethical risk could enter your specific project — your data, model, and deployment, not a generic checklist. Ask: what data are we using and who's missing from it? Who is affected, directly and indirectly? What happens if the model is wrong?	Walk through the 6 Responsible AI principles (Session 4) as a checklist against your specific idea.
2 · Assess	For each risk, judge two things: how likely is it, and how severe would the consequence be? Not every risk deserves equal attention or the same response.	Score each risk on a Likelihood × Severity grid to decide what to fix now versus what to document.
3 · Mitigate	Design specific safeguards before you build further — built into the product, not bolted on after a judge or user finds the problem.	Confidence flags for thin-data areas; community reporting partnerships; accuracy testing specifically in underserved areas (see the Mitigation Toolbox below).
4 · Monitor	Ethical risk doesn't end at launch — real-world conditions and affected people both change. Commit to checking after deployment, not just at demo day.	Track false-alarm and missed-alert rates by neighbourhood over time; fixed review checkpoints; a channel for communities to report problems.

3. Full Worked Example — Team Tropika, Step by Step

The same team and flash-flood prototype from Session 2, now walked through the framework in real depth. Every step ends with a concrete deliverable a judge can inspect.

Step 1 · Identify — five named risks, in plain language

1. **Data coverage gap:** 3 official rain gauges cluster in central Shah Alam; kampung areas 5 km out have none.
2. **Fairness:** if the model is less accurate exactly where there's no data, it silently under-serves the households most likely to flood.
3. **Consent:** the WhatsApp number list needs a real opt-in — not an assumed or scraped list.
4. **Explainability:** residents see a risk label but no reason why the model produced it.
5. **Accountability:** no one is yet named as owner if a false negative causes real harm.

Deliverable: a written list of named risks, each tied to a specific part of the system.

Step 2 · Assess — the Likelihood × Severity grid

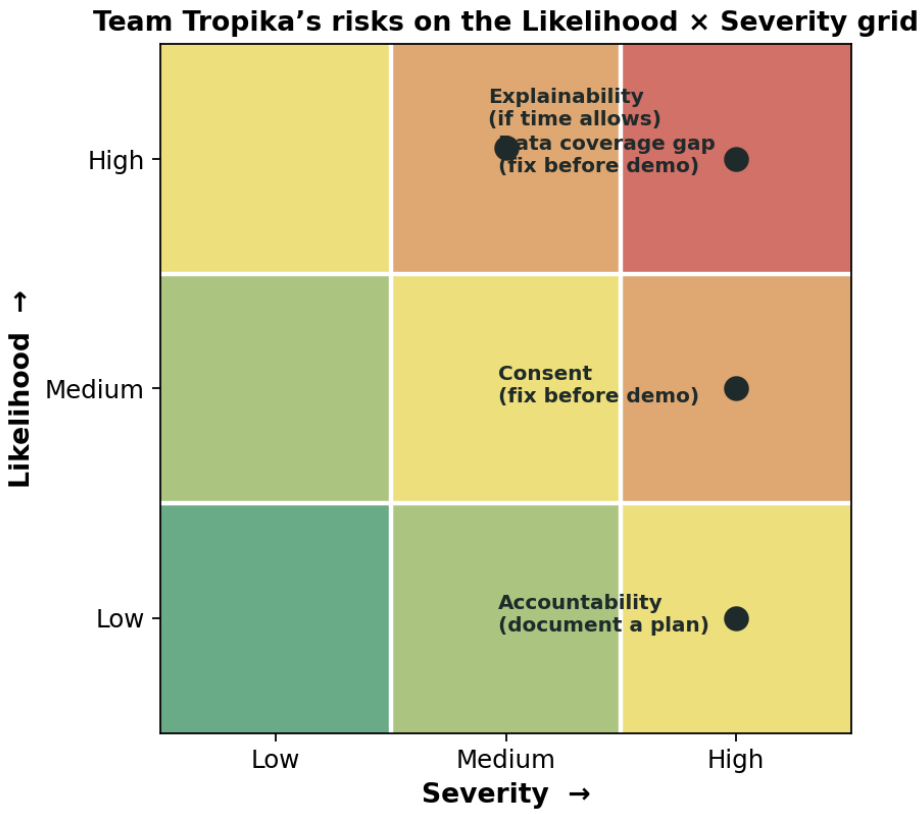


Figure 5.2 — Team Tropika's four headline risks plotted on the grid. Position determines the response: fix, defer, or document.

Risk	Likelihood	Severity	Priority
Data coverage gap	High	High	Fix before demo
Consent	Medium	High	Fix before demo
Explainability	High	Medium	If time allows
Accountability	Low	High	Document a plan

Why this order: the data gap and consent issues both score High severity and are realistic to fix within the weekend, so they get the team's remaining hours. Accountability is High severity too, but as a Low-likelihood, harder-to-fix risk, it gets a written plan instead of a rebuild.

Deliverable: a completed grid the team can show a judge to explain their priorities.

Step 3 · Mitigate — built in, not bolted on

- **Data coverage gap:** a "Data Confidence: Low / Medium / High" tag added to every alert, computed from distance to the nearest gauge. Kampung-area alerts add: "Limited local data — please also check with neighbours."
- **Consent:** the assumed phone list is replaced with an explicit opt-in — residents text "JOIN" to subscribe, with a one-line notice on what data is used and how to opt out.
- **Explainability (bonus):** time allowed one more fix — each alert now shows its top driver, e.g. "Based on: rainfall in your area (68 mm/6 hr)."

Deliverable: an updated prototype plus a one-paragraph "mitigation log" the team can show a judge.

Step 4 · Monitor — a plan for after demo day

A prototype cannot collect real deployment data over a weekend — so the team writes a credible plan instead:

1. **Metric to track:** false-negative rate broken out by distance-to-gauge — not just overall accuracy.

2. **Review checkpoint:** re-check performance after the first real storm event, before any wider rollout.
3. **Feedback loop:** the bot itself collects data — "Reply MISSED if you flooded without an alert."
4. **Named owner:** one team member (or role, in a real deployment) is named as who reviews this data and can adjust thresholds.

Deliverable: a monitoring-plan paragraph added to the final pitch — showing judges the team thought past demo day.

4. The Mitigation Toolbox — Standard Fixes by Challenge

When your team identifies a risk in Activity 5, reach for the matching column below. These are the industry-standard mitigation families.

Challenge	Standard mitigations
Data bias	Audit dataset coverage per group and per area; add local data partners or community reporting; evaluate accuracy disaggregated by group/area, never only in aggregate; document known gaps openly.
Privacy	Data minimisation (collect only what the task needs); aggregation to the coarsest workable resolution; k-anonymity or differential privacy before sharing; explicit opt-in consent with a clear opt-out.
Explainability	Prefer interpretable models where possible; apply SHAP / feature-importance tools to complex ones; surface the top driver of each prediction in plain language to the end user.
Environmental cost	Right-size the model to the problem; start from pretrained models instead of training from scratch; measure and report compute used; question whether a heuristic would do.
Misinformation	Record data provenance (where every dataset came from); cross-check imagery against a second independent source; version and checksum your data so tampering is detectable.

5. Hands-On Activity 5 — Ethical Risk Mapping (10 minutes)

Task: in teams — or on paper if self-studying — identify risks, impacts, and mitigation measures for your project. Example chain: Biased Training Data → Inaccurate Prediction → More Harm to Vulnerable Groups.

Deliverable: a Risk–Mitigation Matrix (Ethical Risk Register). In professional AI governance this exact artefact — a living risk register with named owners — is standard practice, so treat this as a real skill, not a classroom exercise.

Risk identified	Potential impact	Mitigation measure	Owner
e.g. Biased training data	e.g. Inaccurate predictions for underrepresented areas	e.g. Audit dataset coverage; add local data partners	Data lead
Your answer:			

Glossary

Term	Meaning
Gender Shades	The 2018 MIT study exposing large accuracy gaps in facial recognition across skin tone and gender.

Term	Meaning
GAN	Generative Adversarial Network — an AI architecture that generates realistic synthetic images.
Deepfake geography	AI-fabricated satellite imagery that looks authentic.
Likelihood × Severity grid	A prioritisation matrix scoring each risk by probability and consequence.
Risk register	A living document listing risks, impacts, mitigations, and named owners.
Disaggregated evaluation	Measuring model accuracy separately per group or area, not just overall.
Provenance	The documented origin and chain of custody of a dataset.
False negative	A miss — the model fails to flag an event that actually happens (the most dangerous error for warning systems).
Saliency map	A visualisation of which image regions a vision model focused on — how the ruler flaw was caught.

Self-Check Questions

1. The Gender Shades error gap was 0.8% vs 34.7%. Explain how the same mechanism could appear in a land-cover model for Borneo.
2. The skin-cancer model had excellent test accuracy. Why was it still dangerous, and which tool family would have exposed the problem?
3. A risk scores Low likelihood but High severity, and cannot be fixed this weekend. According to the framework, what is the correct response?
4. Why does Team Tropika track false negatives by distance-to-gauge rather than overall accuracy?
5. Pick any project idea and write one row of a risk register for it: risk → impact → mitigation → owner.

Key Takeaways

- GeoAI's five headline challenges — data bias, privacy/surveillance, explainability, environmental cost, and misinformation — all have documented, expensive real-world precedents.
- The framework is a loop: Identify → Assess → Mitigate → Monitor, feeding back into a fresh Identify.
- A Likelihood × Severity grid turns a scary risk list into a realistic weekend plan: fix, defer, or document.
- Good mitigations are built into the product (confidence tags, opt-ins, plain-language explanations) — not bolted on after someone finds the problem.
- Every risk needs a named owner; "the algorithm did it" is never the end of the conversation.

Sources & Further Reading

- Buolamwini & Gebru, "Gender Shades", FAT* (2018); Dutch DPA and Forbes — Clearview AI fines (2024).
- PMC — AI in Dermatology (2022); Zhao et al., "Deep Fake Geography", Cartography & GIS (2021).
- IEA — Energy and AI projections (2025); Climate Impact Partners.
- MDPI — An Operational Ethical Framework for GeoAI (2026); Harvard CGA — Critical Challenges of GeoAI.